



COURSEWORK SPECIFICATION

COMM040 - Text Mining and Natural Language Processing

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Information Extraction

Explore Information Extraction as a pipeline by extracting structured information from U.S. presidential inaugural speeches and analysing it across time.

Environment Setup

1.
 - available files
 - metadata encoded in filenames (year, speaker)
 2. Questions
 - What information is already structured?
 - What information is only available in the text?
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Define the IE Schema

1. Design an IE schema to capture political events and actors mentioned in the speeches.
 2. Example fields:
 - Speaker
 - Year
 - Entity (e.g. country, institution)
 - Action (verb)
 - Topic (optional)
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Entity Extraction

- Using spaCy, apply extraction to multiple speeches.
 - Questions:
 - Which entity types dominate the corpus?
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Event Approximation via Syntax

1. Using dependency parsing:
 - a. Identify main verbs in each sentence.
 - b. Extract simple tuples: (subject, verb, object)
2. Filter tuples where: subject or object is a named entity
3. Questions

- a. Which relations are easy to extract?
 - b. Which political actions are frequent?
 - c. Where does syntax help, and where does it fail?
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Visualisation as IE Analysis

1. Using the extracted structured data, create the following visualisation:
 - a. frequency of entity types over time
 - b. most frequent entities per decade
 - c. Question: What has changed over time?
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